WORKED SOLUTION

STEP II 2011, Question 2

Pure • Diophantine Equations

> **STEP II 2011, Question 2**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/11-s2-q2

The key idea

Whenever you see $x^3 + y^3$, factorise it: $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$. Once $x + y$ is pinned down (to k in (i), to z^2 in (ii)), the factor $x + y$ cancels against part of kz^3, leaving a quadratic condition. Completing the square in that quadratic turns it into "something $= 3\times$ a perfect square", and bounds then leave only a handful of cubes to try. That is why the question asks for the cubes up to 10^3 first.

For reference: $1, 8, 27, 64, 125, 216, 343, 512, 729, 1000$.

Solution

Part (i)

Since $x + y = k$,

$$kz^3 = x^3 + y^3 = (x + y)(x^2 - xy + y^2) = k(x^2 - xy + y^2),$$

and $k > 0$, so $z^3 = x^2 - xy + y^2$. Substituting $y = k - x$:

$$\begin{aligned} z^3 &= x^2 - x(k - x) + (k - x)^2 \\ &= x^2 - kx + x^2 + k^2 - 2kx + x^2 \\ &= k^2 - 3kx + 3x^2, \end{aligned}$$

as required.

The perfect square. Multiply by 4 and complete the square:

$$4z^3 - k^2 = 12x^2 - 12kx + 3k^2 = 3(4x^2 - 4kx + k^2) = 3(k - 2x)^2.$$

So $\dfrac{4z^3 - k^2}{3} = (k - 2x)^2$, the square of the integer $k - 2x$.

The bounds. From the line above, $4z^3 = k^2 + 3(k-2x)^2 \geq k^2$, so $\frac{1}{4}k^2 \leq z^3$. For the upper bound, note $3kx - 3x^2 = 3x(k-x) = 3xy$, so

$$z^3 = k^2 - 3xy < k^2,$$

since x and y are positive. Hence $\frac{1}{4}k^2 \leq z^3 < k^2$.

The case $k = 20$. We look for a solution with $x + y = 20$. The bounds give $100 \leq z^3 < 400$, so from the list of cubes $z^3 \in \{125, 216, 343\}$. Now test whether $(4z^3 - 400)/3$ is a perfect square:

- $z = 5$: $(500 - 400)/3 = 100/3$, not an integer;
- $z = 6$: $(864 - 400)/3 = 464/3$, not an integer;
- $z = 7$: $(1372 - 400)/3 = 324 = 18^2$. ✓

So $(k - 2x)^2 = 18^2$ with $k = 20$. Since $x < y$ means $2x < x + y = k$, we have $k - 2x > 0$, so $k - 2x = 18$, giving $x = 1$ and $y = 19$. Check: $1^3 + 19^3 = 1 + 6859 = 6860 = 20 \times 343 = 20 \times 7^3$. So

$$x = 1, \quad y = 19, \quad z = 7$$

is a solution of $(*)$ with $k = 20$.

Part (ii)

Now suppose $x + y = z^2$. Then

$$kz^3 = (x+y)(x^2 - xy + y^2) = z^2(x^2 - xy + y^2),$$

and dividing by z^2 (allowed, as $z > 0$) gives $x^2 - xy + y^2 = kz$.

> **Ask yourself** How can we get the same "perfect square" trick to work here?

Use the identity $4(x^2 - xy + y^2) = (x+y)^2 + 3(x-y)^2$, which is what we found in (i) in disguise (expand the right-hand side to check). With $x + y = z^2$ and $k = 19$ this becomes

$$76z = z^4 + 3(y-x)^2, \qquad \text{that is} \qquad 3(y-x)^2 = z\,(76 - z^3).$$

Since $y > x$, the left-hand side is positive, so $z^3 < 76$ and therefore $z \in \{1, 2, 3, 4\}$. We need $z(76 - z^3)/3$ to be a perfect square:

- $z = 1$: $75/3 = 25 = 5^2$, so $y - x = 5$; but $x + y = 1$, impossible for positive integers.
- $z = 2$: $2 \times 68/3 = 136/3$, not an integer.
- $z = 3$: $3 \times 49/3 = 49 = 7^2$, so $y - x = 7$ and $x + y = 9$, giving $x = 1$, $y = 8$.
- $z = 4$: $4 \times 12/3 = 16 = 4^2$, so $y - x = 4$ and $x + y = 16$, giving $x = 6$, $y = 10$.

Checking: $1^3 + 8^3 = 513 = 19 \times 27 = 19 \times 3^3$, and $6^3 + 10^3 = 1216 = 19 \times 64 = 19 \times 4^3$. So two solutions of $(*)$ with $k = 19$ are

$$(x, y, z) = (1, 8, 3) \quad \text{and} \quad (x, y, z) = (6, 10, 4).$$

(In fact the argument shows these are the only ones with $x + y = z^2$.)

Checks

- Test the identity of (i) on a small case: $k = 5$, $x = 2$, $y = 3$ gives $x^3 + y^3 = 35 = 5 \times 7$, and $k^2 - 3kx + 3x^2 = 25 - 30 + 12 = 7$. ✓
- Substitute every answer back into $(*)$, as done above: $6860 = 20 \times 343$, $513 = 19 \times 27$, $1216 = 19 \times 64$.
- A computer search over all $x < y$ with $x + y = 20$, and over all $x < y$ with $x + y = z^2$ for $k = 19$, finds exactly these solutions and no others.

Common slips

- Forgetting the sign: $(k - 2x)^2 = 18^2$ also allows $k - 2x = -18$, but that gives $x = 19 > y$, which the condition $x < y$ rules out.
- In (ii), keeping $z = 1$ as a solution: $y - x = 5$ is fine, but it can't coexist with $x + y = 1$.

Beyond STEP

Equations like $x^3 + y^3 = kz^3$ ask which integers k are sums of two rational cubes, a question that leads straight to elliptic curves and is still not fully solved; the famous taxicab number $1729 = 1^3 + 12^3 = 9^3 + 10^3$ lives in the same world. The tricks here (factorising, completing the square, then bounding to get a finite search) are the bread and butter of elementary number theory. Davenport's *The Higher Arithmetic* is a lovely next step.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.